



Lab Report – EE-170L Computer Programming (Lab 10)

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Objectives

- Understand **function call by value**.
- Understand **function call by reference**.
- Learn about **default parameter values** in functions.
- Apply these concepts in practical C++ programs.

Theory

Call by Value: A copy of the variable is passed to the function. Changes inside the function do not affect the original variable.

Call by Reference: The actual memory address is passed. Changes inside the function directly affect the original variable.

Default Parameters: Functions can have default values for parameters, which are used if no argument is provided during the call.

Lab Tasks

Task 1: Function with Parameters

```
#include <iostream>           // include input/output library
using namespace std;         // use standard namespace

void multiply(int num1, int num2) { // function with two integer parameters
    int product = num1 * num2;    // calculate product of num1 and num2
    cout << "Product = " << product; // display result inside function
}
```

```

int main() {                // main function starts
    multiply(6, 7);          // call multiply function with two numbers
    return 0;                // end program
}

```

Task 2: Function with Return Value

```

#include <iostream>          // include input/output library
using namespace std;        // use standard namespace

int getSquare(int number) {  // function with one integer parameter
    int square = number * number; // calculate square of the number
    return square;           // return the result
}

```

```

int main() {                // main function starts
    int result = getSquare(9); // call function and store returned value
    cout << "Square = " << result; // display the square
    return 0;                // end program
}

```

Results

- Task 1 displayed: **Product = 42** (for inputs 6 and 7).
- Task 2 displayed: **Square = 81** (for input 9).

Conclusion

In this lab, I learned:

- *How to pass parameters to functions.*
- *How to return values from functions.*
- *The difference between call by value and call by reference.*
- *How default parameters simplify function calls.*

*Functions make programs **modular, reusable, and efficient.***